



**Politecnico
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Politecnico di Torino

Lab report for the course Qubit Electronics

Master degree in Quantum Engineering

Group 1

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CHAPTER 1

Lab 1

1.1 Introduction

A superconducting qubit can be implemented through an LC circuit whose Hamiltonian is

$$H = \frac{Q^2}{2C} + \frac{\phi^2}{2L}.$$

A parallelism can be traced with an harmonic oscillator where

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2.$$

As a matter of fact, We can rewrite the first Hamiltonian introducing the resonance frequency of the circuit $\omega_{01} = \frac{1}{\sqrt{LC}}$:

$$H = \frac{Q^2}{2C} + \frac{1}{2}C\omega_0^2\phi^2.$$

Formally, it is the same if we substitute $Q \rightarrow p, C \rightarrow m$ and $\phi \rightarrow x$. By using the canonical quantization, we can promote Q and ϕ to (non-commuting) operators. Note that, since they are observables, they are represented by Hermitian operators. The Schrödinger problem can be solved by introducing the ladder operators $a, a^\dagger$, which simplify the expression of the Hamiltonian to

$$\hat{H} = \hbar\omega_{01} \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right).$$

The quantum harmonic oscillator is not a good qubit, since the energy levels are equally spaced and hence we do not have much control over the transitions.

To solve this problem, we substitute the inductance with the Josephson junction. In this way, we introduce a non-linearity which leads to the fact that the energy levels are not equispaced anymore. The Hamiltonian now is

$$\hat{H} = \frac{\hat{Q}^2}{2(C_j + C_s)} - E_j \cos \frac{2\pi\hat{\phi}}{\Phi_0}.$$

The first term represents the **capacitive energy** of the capacitance of the junction C_j and of the shunting capacitance of the resonator C_s . The second term, instead, represents the **inductive energy** due to the Josephson junction. We can rewrite the Hamiltonian in terms of Cooper pairs and account for a voltage bias $V_g = V_0 \cos \omega_{01}t$:

$$\hat{H} = 4E_c(\hat{n} - n_g)^2 - E_j \cos \hat{\phi},$$

where we introduced the charging energy $E_c := \frac{e^2}{2C}$ and the charge number operator $\hat{n} := \hat{Q}/2e$.

Usually, the resonator is implemented through a cavity, whose modes depend on its dimensions. If it is L long, then the allowed wavelengths are

$$\lambda_m = \frac{2L}{m} \quad m \in \mathbb{N}.$$

We will work under the **one mode approximation**, which allows us to treat the resonant cavity as an equivalent LC circuit. Placing the qubit near the resonator allows us to achieve coupling between the two.

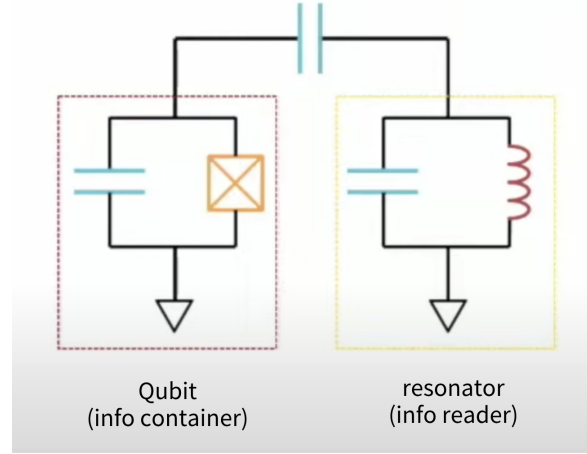


Figure 1.1: Circuitual model of the coupling between the resonator and the qubit.

Based on the strength of the coupling, the resonance frequency depends on the state of the qubit.

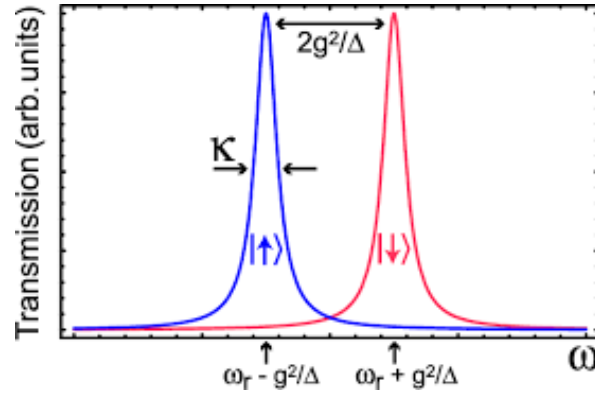


Figure 1.2: Frequency shift due to the coupling with the qubit

A microwave signal, at a frequency close to the resonance one, is sent to the resonator. Depending on how it is shifted, we are able to perform a readout of the qubit state.

1.2 RLC parallel circuit analysis

The ratio between the voltage at the upper node and the current flowing in the current source represents the impedance seen by the generator I_1 . As it can be seen by figure 1.4, it is frequency-dependent.

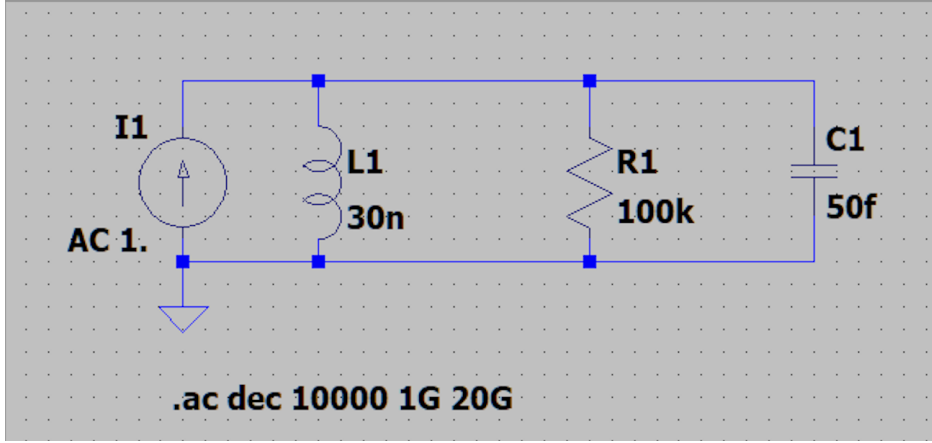


Figure 1.3: RLC circuit

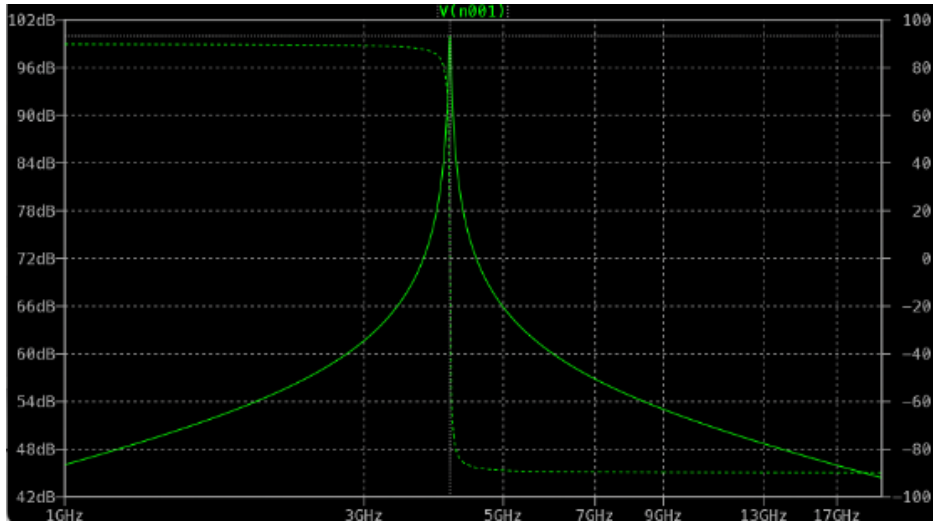


Figure 1.4: Impedance seen from the current source

The values of the resonance frequency and of the quality factor predicted by the theory and the ones resulting from the simulation are reported below:

$$\begin{aligned} f_{th} &= 4.1094 \text{ GHz}, & f_{sim} &= 4.1096 \text{ GHz}, \\ Q_{th} &= 1.2910 \cdot 10^2, & Q_{sim} &= 1.2773 \cdot 10^2. \end{aligned}$$

Both quantities are in good agreement with their theoretical predictions.

1.3 Analysis of RLC parallel circuit changing the R,L and C values

By varying the values of the inductance, the resistance and of the capacitance, we obtained the following results.

It is evident that the resonance frequency obtained from the simulation does not depend on the resistance, but only on the reactive elements. This is in line with what we expected from the theoretical formula. On the other hand, the quality factor does depend on the resistance linearly. This is clear

L (H)	R (Ω)	C (F)	f_{th} (Hz)	Q_{th}	f_{sim} (Hz)	Q_{sim}
3.00E-08	2.00E+05	5.00E-14	4.11E+09	2.58E+02	4.11E+09	2.71E+02
3.00E-08	5.00E+04	5.00E-14	4.11E+09	6.45E+01	4.11E+09	6.39E+01
6.00E-08	1.00E+05	5.00E-14	2.91E+09	9.13E+01	2.91E+09	9.24E+01
1.50E-08	1.00E+05	5.00E-14	5.81E+09	1.83E+02	5.81E+09	1.81E+02
3.00E-08	1.00E+05	1.00E-13	2.91E+09	1.83E+02	2.91E+09	1.83E+01
3.00E-08	1.00E+05	2.50E-14	5.81E+09	9.13E+01	5.81E+09	9.21E+01

Table 1.1: Analysis of RLC Parallel Circuit

by looking at the first two rows of the table. Overall, the results are consistent with their expected values.

1.4 Design an RLC parallel circuit

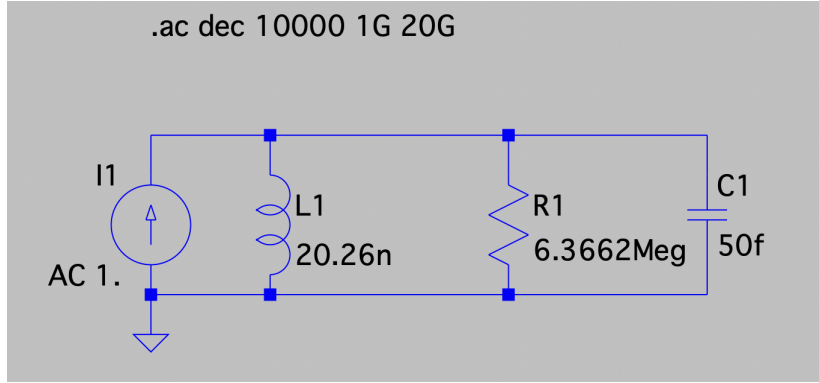


Figure 1.5: Design of RLC circuit

With the given constraints on the capacitance, the resonance frequency and on the quality factor, namely,

$$C = 5 \cdot 10^{-14} \text{ F}, \quad f_R = 5 \text{ GHz}, \quad Q = 10^4,$$

the formulas to compute L and R are:

$$L = \frac{1}{(2\pi f_R)^2 \cdot C}, \quad R = \frac{Q}{2\pi f_R \cdot C}.$$

From which we got:

$$L = 2.0264 \cdot 10^{-8} \text{ H} \quad R = 6.366 \cdot 10^6 \Omega$$

By inserting the computed parameters into the simulation in LTspice, we got a resonance frequency and a quality factor, respectively, of:

$$f_{sim} = 5.00035 \text{ GHz}, \quad Q_{sim} = 9.57 \cdot 10^3,$$

consistent with the setup.

1.5 Josephson Junction

1.5.1 Circuit analysis

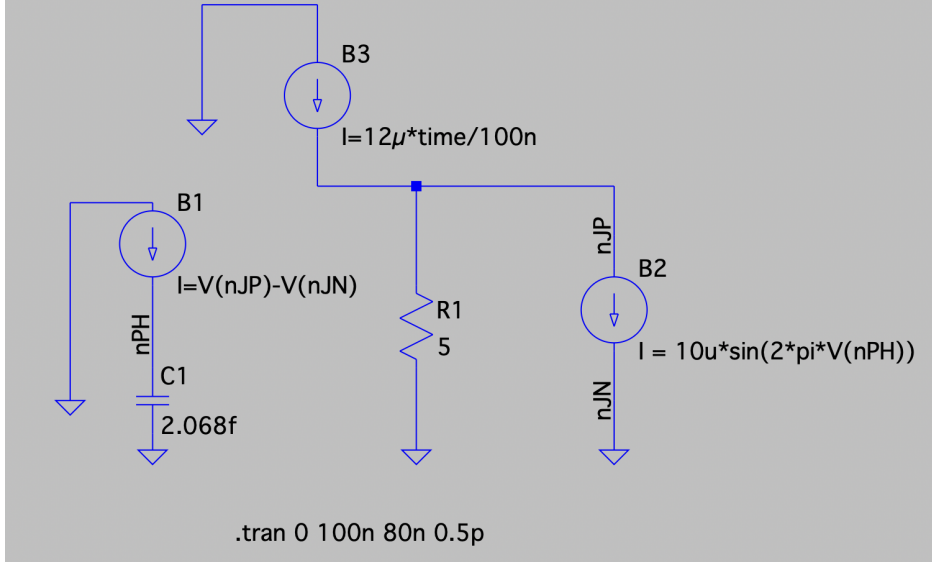


Figure 1.6: Circuitual model of the Josephson junction

The current generator B2 implements the current-phase relation, effectively modeling the current flowing in a Josephson Junction:

$$i = I_c \sin(2\pi\theta(t)) \iff i_{sim} = 10\mu \cdot \sin(2\pi \cdot V(nPH)).$$

The critical current I_c corresponds to $10\mu\text{A}$, whereas the phase difference between the two superconducting banks is governed by the voltage at the node nPH, namely by $V(nPH)$, coherently with the voltage-phase relationship of the JJ:

$$\theta = \frac{2\pi}{\phi_0} \int V(t) dt, \quad (1.1)$$

where $V(t)$ is the voltage difference across the junction.

In our case, $V(nPH)$ is the voltage across the capacitor C_1 and it is equal to:

$$V(nPH) = \frac{1}{C_1} \int I(t) dt = \frac{1}{C_1} \int V(nJP) - \cancel{V(nJN)} dt = \frac{1}{C_1} \int V(nJP) dt. \quad (1.2)$$

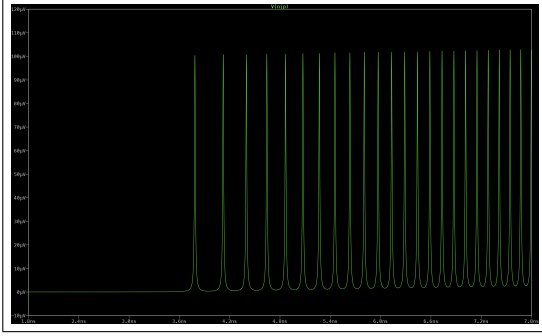
We can observe that the capacitance at the denominator has the same role of the quantity $\frac{1}{\phi_0}$ and thus the two formulas, (1) and (2), are identical from a qualitative point of view. Therefore, the circuit is a good approximation of the behavior of the Josephson Junction.

1.5.2 Results analysis

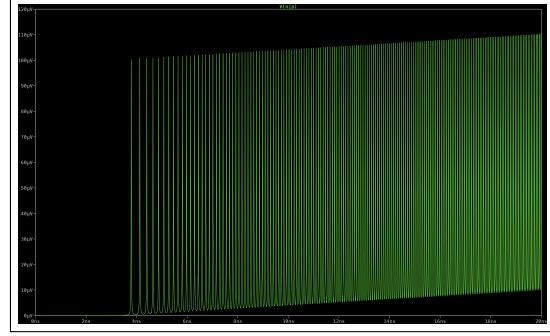
By measuring the voltage at the node nJP, we obtained that the first five peaks occur at the following times:

We observe that the time intervals between each peak decrease progressively. In addition, we notice a general increase of the voltage, while maintaining an oscillating behavior.

	t (s)	Δt (s)
t_1	3.78522	0
t_2	4.12275	0.338
t_3	4.39925	0.277
t_4	4.64325	0.244
t_5	4.86625	0.223



(a) First peaks of the voltage



(b) Increase of the voltage

We can justify the increase in the frequency of the peaks by observing that the voltage $V(\text{nPH})$ has a quadratic dependence on time. As a consequence, the argument of the sine also increases, inducing a shortening of the period. From a mathematical standpoint, the voltage at node nPH is given by:

$$V(\text{nPH}) = \frac{1}{C_1} \int V(\text{nJP}) dt = \frac{1}{C_1} \int (I_{B3} - I_{B2}) \cdot R_1 dt. \quad (1.3)$$

By neglecting the contribution of the current I_{B2} , because of the complicated time dependence, we are still able to derive a good approximation of the behavior of the circuit. I_{B3} , from LTspice, is defined as:

$$I_{B3} = 12\mu \frac{t}{100n} = \alpha t \quad (1.4)$$

where $\alpha = \frac{12\mu}{100n}$. Therefore, from the integration, $V(\text{nPH})$ results:

$$V(\text{nPH}) = \frac{\alpha}{C_1} \frac{t^2}{2}. \quad (1.5)$$

So, the current flowing in the JJ is:

$$i = I_c \sin \left(2\pi \frac{\alpha}{C_1} \frac{t^2}{2} \right). \quad (1.6)$$

The general increase of the voltage is due to the contribution of both the oscillating current of the JJ, namely (1.6), and the increasing driving current $I_{B3} \propto t$.

In conclusion, the qualitative behavior of the voltage is

$$V(t) \propto t + \sin(t^2). \quad (1.7)$$

1.6 Analysis of the circuit - Qubit coupled to a resonator

By varying the inductance of the qubit with values of 20.45nH, 23.45nH and 26.45nH, we observe a shift of the peaks in the response of all the elements. They are reported respectively in red, in teal and in green.

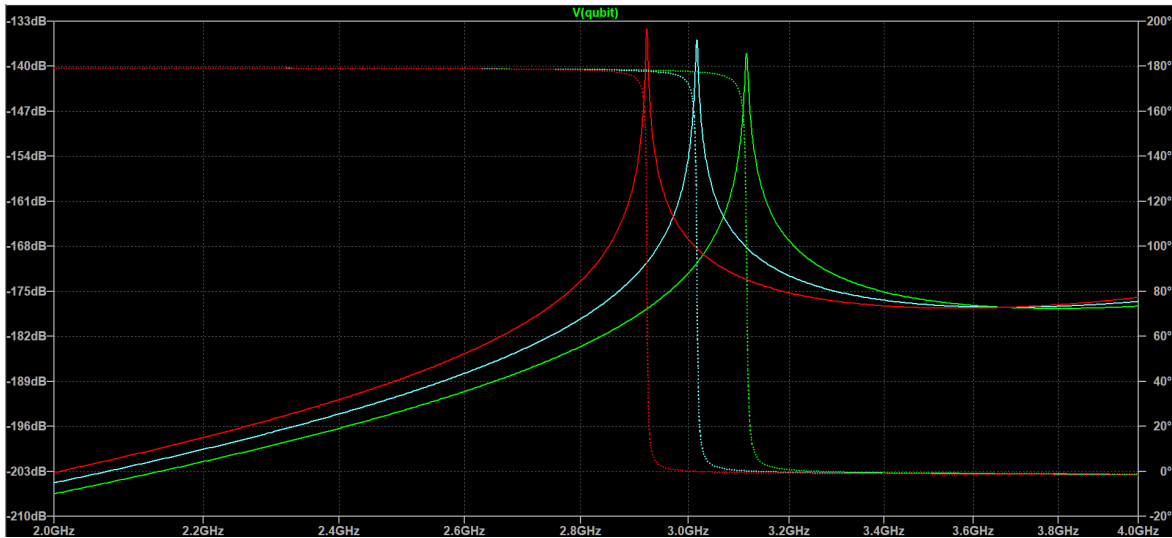


Figure 1.8: Response of the qubit

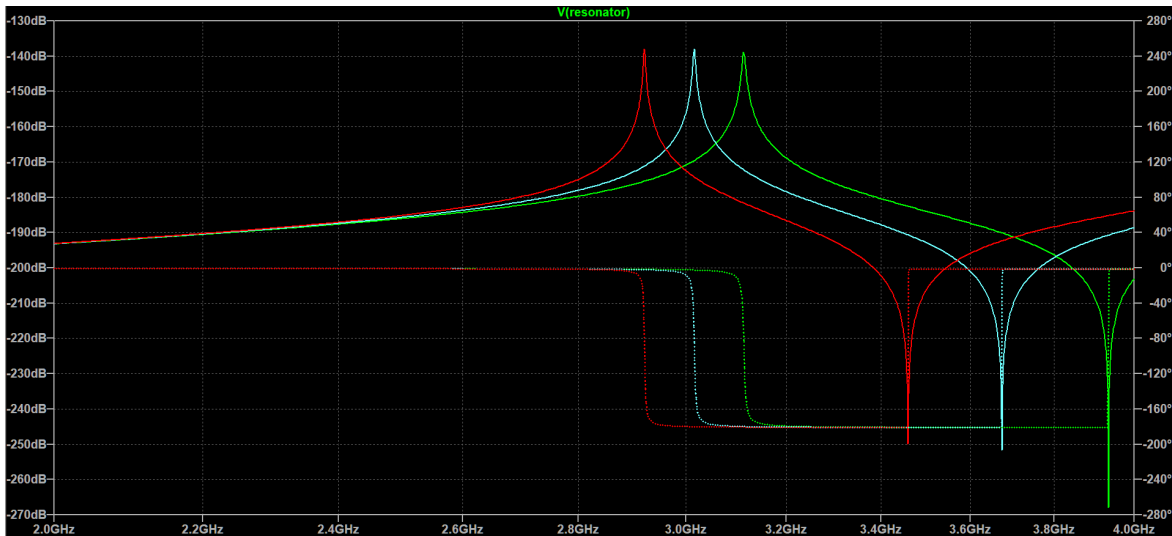


Figure 1.9: Response of the resonator

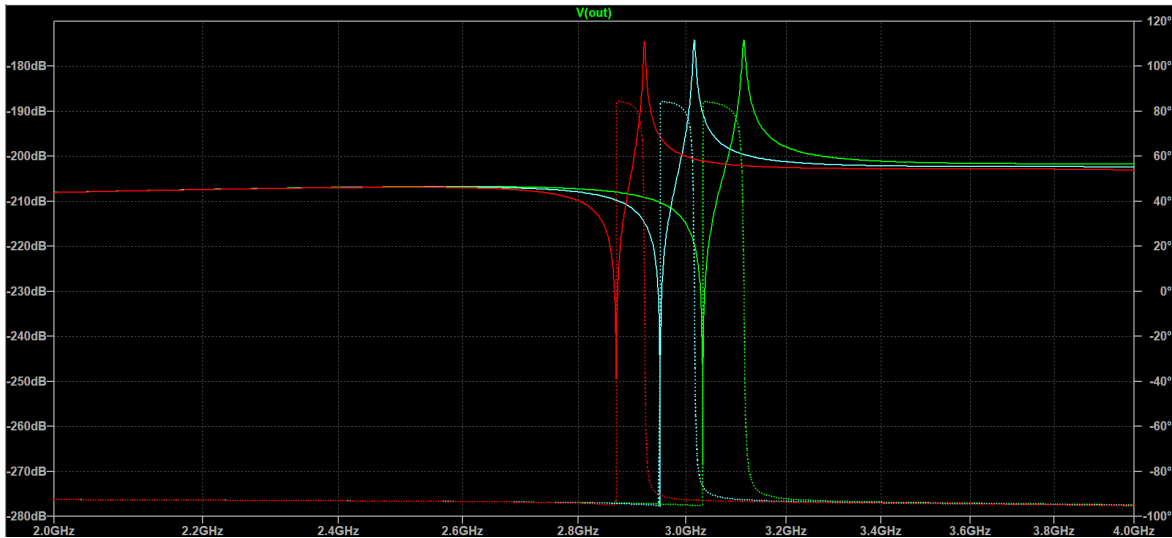


Figure 1.10: Response at the output

Inductance (H)	Peak 1 (Hz)
20.45E-09	3.114E+09
23.45E-09	3.016E+09
26.45E-09	2.922E+09

As stated in the introduction 1.1, the resonance frequency of the qubit has an influence on the resonance frequency of the resonator. From a physical point of view, it can be traced back to a change in the state of the qubit.